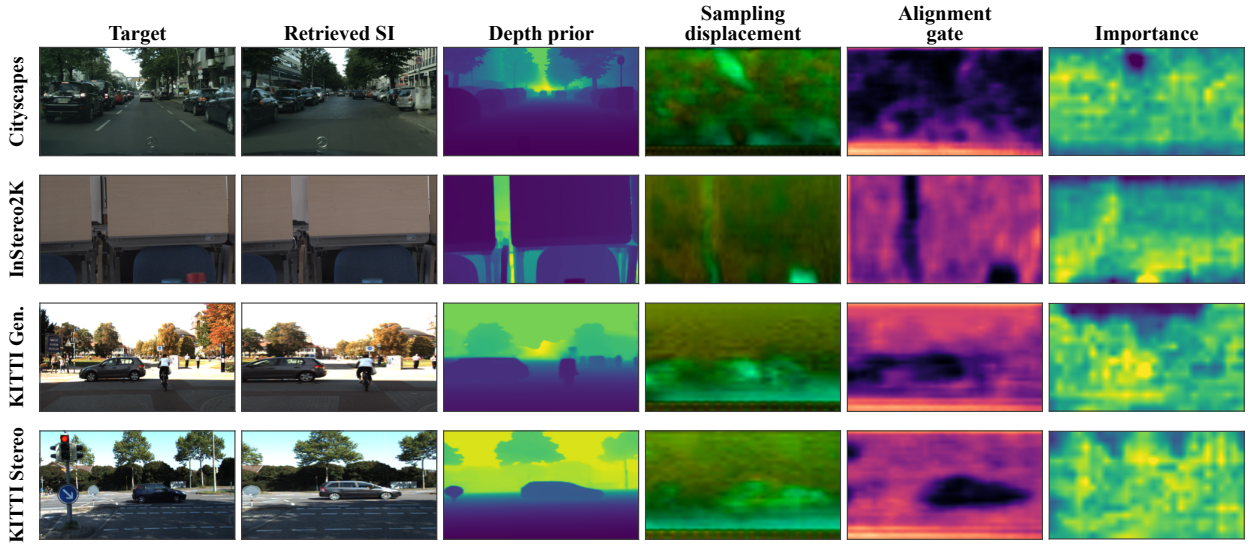


Fig. 1: Visualization of geometric alignment and importance modulation across the four test datasets. From left to right, each row shows the target image, retrieved side information (SI), depth prior, learned sampling displacement, alignment gate c_f, and encoder-predicted latent retention response $1-\eta M_{\text{pred}}$. The sampling displacement represents the learned sampling offsets.

I. Additional Experimental Analysis

A. Visualization of Alignment and Importance Modulation

Fig. 1 visualizes the intermediate responses of DAGA and the importance branch on the four test datasets. The sampling displacement and alignment gate characterize the processing of retrieved SI. The response $1-\eta M_{\text{pred}}$ shows the latent retention predicted from the target image.

A consistent pattern across all four examples is that the learned displacement varies spatially with scene geometry. Instead, the sampling displacement varies spatially and is shaped by scene geometry. The alignment gate c_f further suppresses regions whose content is less reliable after warping, while preserving regions with stronger target-reference consistency.

In the first row from Cityscapes, the retrieved SI is weakly related to the target. Large appearance differences are visible on both sides of the street scene. Correspondingly, the alignment gate assigns low responses to a large portion of the image, indicating limited confidence in the warped SI. The retention response remains relatively high over broad regions, which means that more target latent information is preserved before quantization when the retrieved SI is less reliable.

In the second row from InStereo2K, the retrieved SI is well matched to the target except for narrow gaps and a local object difference near the lower central-right area. The alignment gate mainly suppresses these inconsistent regions, while remaining smoother on the large shared planar areas. The retention response is lower on the flat and well-supported regions, but higher around local details and less reliable areas, which is consistent with SI-aware rate allocation.

The third row from KITTI General shows a related scene with noticeable differences in the vehicle region and several background details. The alignment gate is correspondingly reduced around the moving car and other mismatched structures, while smoother background regions receive higher confidence. The retention response preserves more target information around detailed or weakly matched regions and attenuates it more in spatially consistent areas.

The fourth row from KITTI Stereo follows the same trend. The vehicle region and the left image boundary exhibit stronger mismatch, and the gate is reduced in these locations. In contrast, the retention response highlights detailed structures such as trees and object boundaries, while assigning lower retention to more homogeneous regions such as the sky. This suggests that the encoder reserves more bits for regions that are either visually detailed or less recoverable from SI.

Overall, Fig. 1 illustrates two complementary roles. The alignment gate is a decoder-side signal that reflects the local reliability of retrieved SI after geometric alignment. The retention response is an encoder-side signal predicted from the target alone, which controls how much target information is preserved before quantization. Their spatial patterns are related but not identical, indicating that RADIC does not simply trust retrieved SI everywhere. Instead, it selectively uses SI where alignment

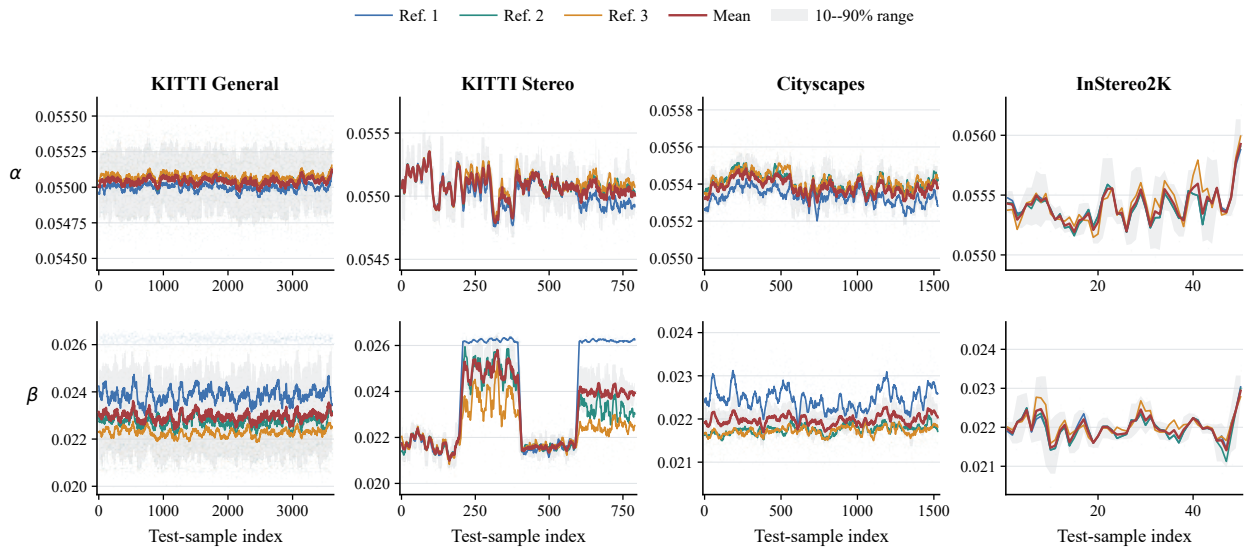


Fig. 2: Sample-wise variation of the DAGA affine parameters α and β over the complete test sets. α controls the depth-dependent slope of the coarse horizontal displacement, whereas β represents its depth-independent offset. Colored curves show rolling means for the three retrieved references, the red curve denotes their mean, faint points show individual samples, and the gray region indicates the rolling 10–90 percentile range. The horizontal axis represents test samples.

is reliable and retains more target information where SI support is weak or uncertain.

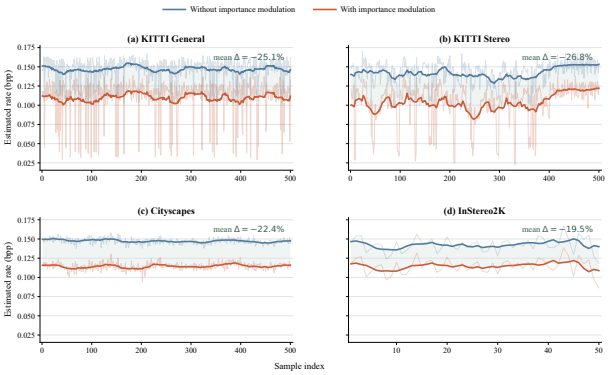


Fig. 3: Sample-wise effect of SI-aware importance modulation on the estimated coding rate. Thin curves show per-image rates, thick curves show moving averages, and the shaded region denotes the rate reduction. KITTI General, KITTI Stereo, and Cityscapes use 500 targets, while InStereo2K uses all 50 test targets.

B. Diagnostics for SI Utilization and Rate Allocation

a) Effect of Importance Modulation on Rate: We evaluate the rate effect of the importance branch by comparing the learned retention response $1 - 0.7M_{\text{pred}}$ with $M_{\text{pred}} = 0$. As shown in Fig. 3 and Table II, learned

TABLE I
FEATURE ALIGNMENT RESULTS OF DAGA.

Dataset	L1 before ↓	L1 after ↓	Reduction (%) ↑	Cosine gain ↑
KITTI General	0.5322	0.4498	15.47	+0.0089
KITTI Stereo	0.5306	0.4492	15.34	+0.0092
Cityscapes	0.5101	0.4288	15.93	+0.0147
InStereo2K	0.5183	0.4329	16.47	+0.0137

retention reduces the mean estimated bitrate by 19.50–26.83% across the four datasets. The result shows that the importance branch directly affects the coding rate.

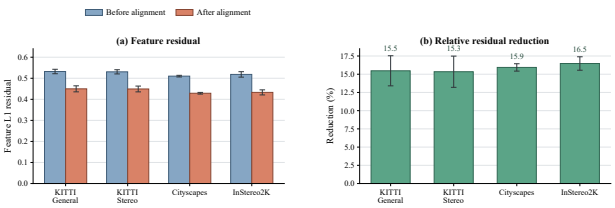


Fig. 4: Internal feature-alignment diagnostics for DAGA. Left: mean feature L1 residual before and after geometric sampling. Right: relative residual reduction over target-reference pairs. KITTI General, KITTI Stereo, and Cityscapes use 500 targets with three references each; InStereo2K uses all 50 targets. Error bars show standard deviations over target-reference pairs.

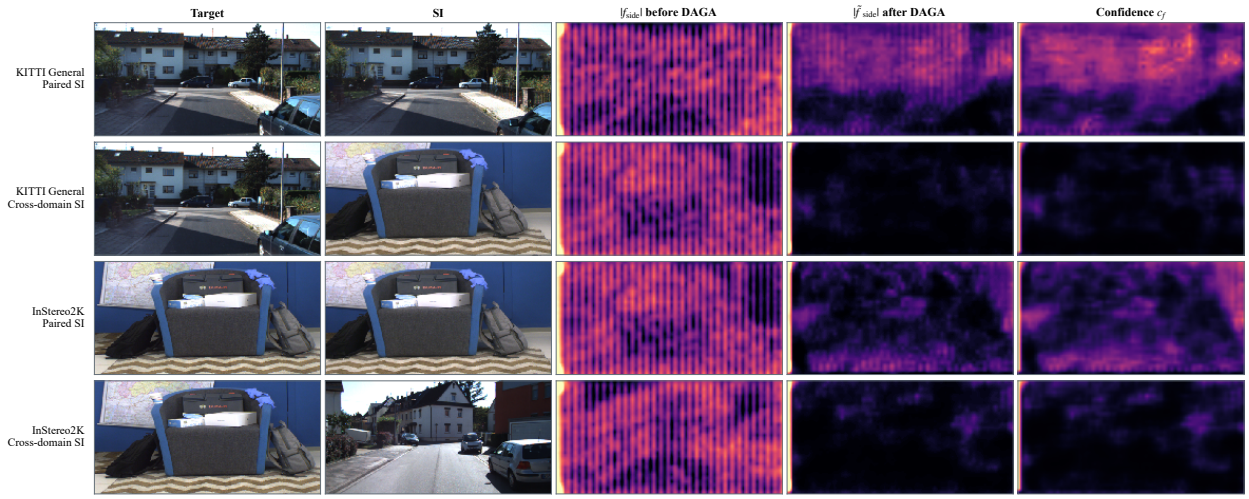


Fig. 5: Visualization of DAGA under paired and cross-domain SI ($K = 1$). Each row shows the target image, retrieved SI, channel-averaged absolute reference features before and after DAGA, and the learned confidence map c_f. DAGA preserves informative features from paired SI while suppressing less relevant features under cross-domain SI. Feature responses are visualized using an asinh transformation with independent normalization.

C. Analysis of DAGA Parameters

The coarse horizontal displacement is defined as

$$flow_x^{\text{coarse}} = \alpha d_{inv} + \beta, \tag{1}$$

where α controls the depth-dependent slope and β represents the depth-independent global offset. Fig. 2 plots their sample-wise values over the complete test sets.

The α curves remain tightly clustered on KITTI General, KITTI Stereo, and Cityscapes, with only limited variation across the three retrieved references. This indicates that the depth-conditioned component of the coarse alignment is relatively stable and is not dominated by a particular reference. In other words, DAGA uses the depth prior in a consistent manner across different outdoor datasets.

By contrast, the β values show larger sample-wise variation, especially on KITTI Stereo. Differences among the three references are also more visible for β than for α. This behavior reflects the stronger sample dependence of the global displacement term.

InStereo2K exhibits larger fluctuations in both α and β, which is consistent with its indoor scenes, shorter depth ranges, and more abrupt depth discontinuities. Even in this case, the three references still follow similar trends, indicating that the learned affine component remains coherent.

These results characterize the two terms used in the coarse displacement. The depth-dependent term varies within a relatively narrow range on most datasets, while the global offset shows stronger sample-wise variation. The residual flow subsequently refines this coarse displacement.

TABLE II
ESTIMATED BITRATE WITH UNIT AND LEARNED LATENT RETENTION.

Dataset	Samples	Unit retention (bpp)	Learned retention (bpp)	Reduction (%)
KITTI General	500	0.14686	0.10995	25.13
KITTI Stereo	500	0.14223	0.10407	26.83
Cityscapes	500	0.14740	0.11443	22.37
InStereo2K	50	0.14226	0.11452	19.50

a) Feature-Space Effect of DAGA.: As shown in Fig. 4 and Table I, we measure the residual between the target feature and each retrieved feature before and after DAGA sampling to evaluate feature-space compatibility.

Across the four datasets, DAGA reduces the mean L1 residual by 15.34–16.47% and produces a small but consistent increase in cosine similarity. The results show that geometric sampling moves retrieved features closer to the target representation.

Fig. 5 further illustrates the effect of DAGA under different SI correlations with $K = 1$. Under paired SI, the retrieved images share corresponding scene content with the targets, and DAGA preserves structured feature responses through alignment and confidence gating. In contrast, cross-domain SI introduces substantial differences in scene content, resulting in lower confidence and stronger suppression of mismatched reference features. These results show that DAGA enables spatially selective utilization of SI even when only a single reference is available.

D. Runtime Measurement Protocol

RADIC is measured on an NVIDIA L40 GPU with paired SI ($K = 1$), 256×512 inputs, a batch size of one, and two denoising steps. Each depth-processing mode uses ten Cityscapes targets with three repetitions after five warm-up runs. CUDA is synchronized at timing boundaries. RADIC uses FP32, while Depth Anything follows its native inference precision.

Cached depth is computed once by Depth Anything and retained on the device. Online depth is recomputed from the reference image in every timed decoding pass, including depth preprocessing and postprocessing. Reference feature extraction, DAGA, MCA, diffusion and image synthesis are executed in both modes. Entropy-model likelihood evaluation is assigned to the encoder. Actual arithmetic coding, model initialization, input I/O, fixed empty-text conditioning and diagnostic statistics are excluded. Online depth estimation takes 0.0885 ± 0.0066 s; complete decoding takes 0.2157 ± 0.0139 s, compared with 0.1245 ± 0.0003 s for cached depth.

The encoder is additionally measured over ten repetitions of the same ten targets after twenty warm-up runs, giving 20.59 ± 0.69 ms. Importance prediction and latent modulation take 0.186 ± 0.022 ms in synchronized wall time, with 0.109 ± 0.015 ms of GPU execution measured using CUDA events. Bypassing these operations in the same encoder gives 20.49 ± 0.66 ms; this is an encoder ablation using the same RADIC weights.